

1. Create Database

```
CREATE DATABASE flight_db;
```

Use it:

```
USE flight_db;
```

2. Create Main Flight Table (Managed Table)

Based on your dataset columns:

```
CREATE TABLE flights (  
  year INT,  
  month INT,  
  day INT,  
  day_of_week INT,  
  dep_time INT,  
  crs_dep_time INT,  
  arr_time INT,  
  crs_arr_time INT,  
  unique_carrier STRING,  
  flight_num INT,  
  tail_num STRING,  
  actual_elapsed_time INT,  
  crs_elapsed_time INT,  
  air_time INT,  
  arr_delay INT,  
  dep_delay INT,  
  origin STRING,  
  dest STRING,  
  distance INT,  
  taxi_in INT,  
  taxi_out INT,  
  cancelled INT,  
  cancellation_code STRING,  
  diverted INT,  
  carrier_delay STRING,  
  weather_delay STRING,  
  nas_delay STRING,  
  security_delay STRING
```

```
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
STORED AS TEXTFILE;
```

3. Load CSV Data into Hive Table

If CSV is in local Linux filesystem:

```
LOAD DATA LOCAL INPATH '/home/hduser/flights.csv'  
INTO TABLE flights;
```

If CSV is in HDFS:

```
LOAD DATA INPATH '/user/hduser/flights.csv'  
INTO TABLE flights;
```

4. Show Data

```
SELECT * FROM flights LIMIT 10;
```

5. Creating External Hive Table

Assignment asks for external table too.

Example:

```
CREATE EXTERNAL TABLE flights_external (  
  year INT,  
  month INT,  
  day INT,  
  day_of_week INT,  
  dep_time INT,  
  crs_dep_time INT,  
  arr_time INT,  
  crs_arr_time INT,  
  unique_carrier STRING,  
  flight_num INT,  
  tail_num STRING,  
  actual_elapsed_time INT,  
  crs_elapsed_time INT,
```

```
air_time INT,  
arr_delay INT,  
dep_delay INT,  
origin STRING,  
dest STRING,  
distance INT,  
taxi_in INT,  
taxi_out INT,  
cancelled INT,  
cancellation_code STRING,  
diverted INT,  
carrier_delay STRING,  
weather_delay STRING,  
nas_delay STRING,  
security_delay STRING  
)  
ROW FORMAT DELIMITED  
FIELDS TERMINATED BY ','  
LOCATION '/user/hive/flight_external/';
```

Difference:

- Managed table → data deleted if table dropped
 - External table → data remains even if table dropped
-

6. Insert New Values

```
INSERT INTO flights VALUES (  
  2024, 5, 12, 1,  
  1030, 1015,  
  1230, 1215,  
  'AI', 101,  
  'TN1234',  
  120, 120, 100,  
  15, 15,  
  'DEL', 'BOM',  
  1148,  
  10, 10,  
  0, NULL,  
  0,  
  '0','0','0','0'  
);
```

7. Alter Table (Add New Field)

Assignment asks altering table.

Example add airline_name:

```
ALTER TABLE flights
ADD COLUMNS (airline_name STRING);
```

Verify:

```
DESCRIBE flights;
```

8. Create Another Table for Join

Create airline details table:

```
CREATE TABLE airline_info (
  unique_carrier STRING,
  airline_name STRING
)
ROW FORMAT DELIMITED
FIELDS TERMINATED BY ',';
```

Insert values:

```
INSERT INTO airline_info VALUES ('AI', 'Air India');
INSERT INTO airline_info VALUES ('6E', 'IndiGo');
INSERT INTO airline_info VALUES ('SG', 'SpiceJet');
INSERT INTO airline_info VALUES ('UK', 'Vistara');
```

9. Join Tables

```
SELECT
  f.flight_num,
  f.origin,
  f.dest,
  a.airline_name
FROM flights f
JOIN airline_info a
ON f.unique_carrier = a.unique_carrier
LIMIT 20;
```

10. Create Index on Flight Information Table

```
CREATE INDEX idx_flight_num  
ON TABLE flights (flight_num)  
AS 'COMPACT'  
WITH DEFERRED REBUILD;
```

Build index:

```
ALTER INDEX idx_flight_num  
ON flights  
REBUILD;
```

11. Average Departure Delay Per Day

This is your required query:

```
SELECT  
  year,  
  month,  
  day,  
  AVG(dep_delay) AS avg_departure_delay  
FROM flights  
GROUP BY year, month, day  
ORDER BY year, month, day;
```

12. Drop Table / Database

Required by assignment:

Drop table:

```
DROP TABLE airline_info;
```

Drop database:

```
DROP DATABASE flight_db;
```

If not empty:

```
DROP DATABASE flight_db CASCADE;
```

